Notes on Demand Estimation

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Roadmap

1. Random utility discrete choice model
2. Estimation (Berry 1994)
3. Limitations of logit demand model
4. Introduce random coefficients logit / BLP as the workhorse extension

Goal for the lecture: take tools from IO and make them usable for finance settings with differentiated products. Become an intelligent consumer of these models.

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Why Estimate Demand?

Demand estimates allow us to move from observed choices to economic primitives and counterfactual predictions.

They allow us to:

- Estimate price elasticities and substitution patterns
- Measure product quality and brand value
- Quantify firms' market power and markups
- Simulate mergers and other counterfactual policies
- Measure the welfare effects of new products

Examples in Finance

Setting	Choice	Price
Banking	Which bank / deposit account	Deposit spread, fees
Mortgages	Which lender / contract	Interest rate, points, fees
Brokerage	Which broker / platform	Commissions, bid-ask costs
Asset management	Which fund / ETF	Expense ratio, load, performance fees
Insurance	Which policy / insurer	Premium

Random Utility Framework for Discrete Choice

Discrete Choice and Demand

- Each consumer chooses one alternative from a finite choice set (discrete quantities; can be extended to multiple quantities)
- Model a consumer's demand as a multinomial choice problem
- Examples in finance:
  - Choosing a bank (deposits/mortgages)
  - Choosing a broker/making trading decisions
  - Choosing ETFs/bonds/investments
  - Buying a car and making financing decisions

Random Utility Framework for Discrete Choice

Indirect utility of consumer i from consuming product j in market t is:

$$u_{ijt} = \alpha p_{jt} + x_{jt}\beta + \xi_{jt} + \varepsilon_{ijt}$$

where:

- p_{jt} is the price of product j in market t
- x_{jt} is a vector of observable characteristics of product j in market t
- ξ_{jt} is an unobserved product characteristic/demand shock
- ε_{ijt} is a consumer-product-market specific demand shock
- Often allow the pref. to vary across consumers (i.e. α_i, β_i)
- $-(\beta/\alpha)$: value of characteristic in dollar terms

What is the interpretation of the error terms and what purpose do they serve?

Consumer's problem

Consumer's problem is to choose the product in the set of available products $\mathcal{J}$ that maximizes the consumer's utility

$$\max_{j \in \mathcal{J}} u_{ijt}$$

Under the assumption that $\varepsilon_{ijt} \sim T1EV$ (Type-1 Extreme Value), the probability consumer i chooses j is :

$$\Pr(i \,\text{chooses}\ j) = \frac{\exp\left(\alpha p_{jt} + x_{jt}\beta + \xi_{jt}\right)}{\sum_{l \in \mathcal{J}} \exp(\alpha p_{lt} + x_{lt}\beta + \xi_{lt})}$$

The T1EV assumption gives us a closed-form expression for the choice probabilities.

Market Share

Often only have access to aggregate/market-share level data

The market share for product j in market t is then:

$$s_{jt} = \frac{\exp\left(\alpha p_{jt} + x_{jt}\beta + \xi_{jt}\right)}{\sum_{l \in \mathcal{J}} \exp(\alpha p_{lt} + x_{lt}\beta + \xi_{lt})}$$

Need to normalize the level of utility

- Typically normalize the utility of the outside good to zero
- Utility is then defined relative to the outside good

Estimation

Note that if we write shares in logs we have

$$\ln(s_{jt}) = \alpha p_{jt} + x_{jt}\beta + \xi_{jt} - \ln\left(\sum_{l\in\mathcal{J}} \exp(\alpha p_{lt} + x_{lt}\beta + \xi_{lt})\right)$$

Given that the utility of the outside good is normalized to zero, we have

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha p_{jt} + x_{jt}\beta + \xi_{jt}$$

Estimation

Can estimate the model using OLS (Berry 1994)

$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha p_{jt} + x_{jt}\beta + \xi_{jt}$$

- Dependent variable: $\ln(s_{jt}) - \ln(s_{0t})$
- Independent variables: $[p_{jt}, x_{jt}]$

Note: could also estimate

$$\ln(s_{jt}) = \alpha p_{jt} + x_{jt}\beta + \xi_{jt} - \ln\left(\sum_{l \in \mathcal{J}} \exp(\alpha p_{lt} + x_{lt}\beta + \xi_{lt})\right)$$

with OLS and include a market fixed effect to absorb the nonlinear term $\left(\ln\left(\sum_{l \in \mathcal{J}} \exp(\alpha p_{lt} + x_{lt}\beta + \xi_{lt})\right)\right)$.

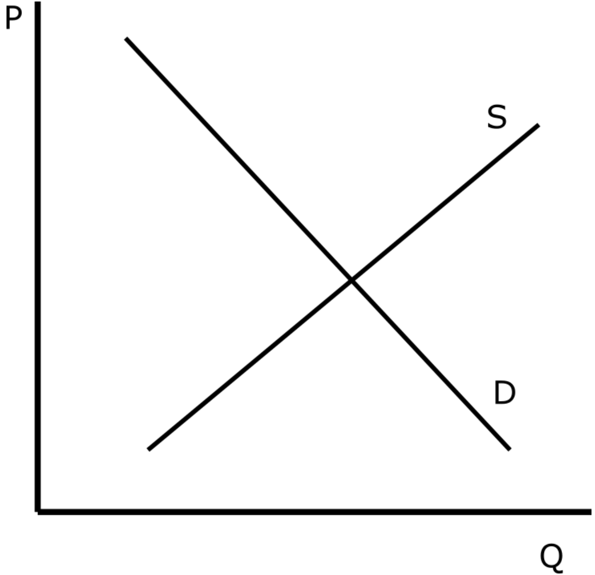
Estimation

- In this regression we are regressing quantities on price
  - Is this demand or supply?

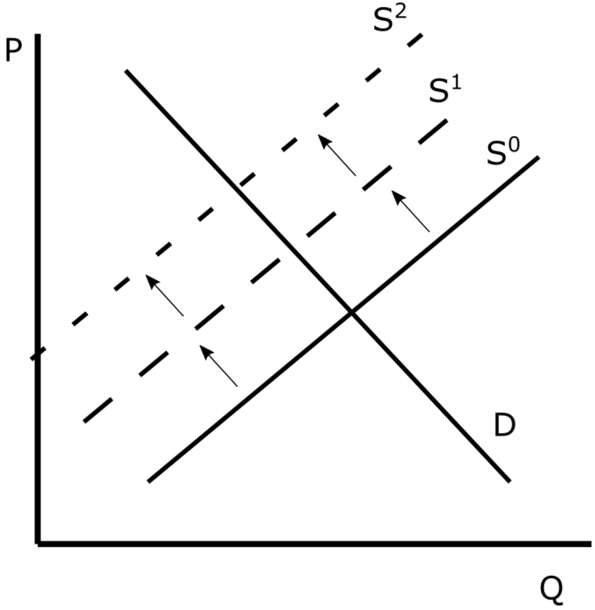
$$\ln(s_{jt}) - \ln(s_{0t}) = \alpha p_{jt} + x_{jt}\beta + \xi_{jt}$$

- If firms observe the demand shocks ξ_{jt} prior to setting prices, then prices will be endogenous
  - If a firm knows ξ_{jt} is high, it will find it optimal to set a higher price p_{jt}
  - Endogeneity of prices will cause OLS estimates of $-\alpha$ to be biased downwards i.e. the estimates will suggest that consumers are less sensitive to prices than they actually are

Estimation



Estimation



Instruments for Prices

Need an instrument for prices that is correlated with prices but uncorrelated with demand shocks/unobserved product characteristics

Cost shifters / supply shifters

- Use variation in marginal costs, funding costs, or regulatory costs
- Intuition: costs affect prices but should not directly affect consumer tastes
- Limitation: often difficult to measure at the product-market level

Hausman instruments

- Use prices of the same product in other markets
- Example: instrument for a product's price in Connecticut with its price in Massachusetts
- Intuition: cost shocks are correlated across markets
- Limitation: other markets may share common demand shocks

Instruments for Prices

BLP instruments

- Use non-price characteristics of own and rival products
- Intuition: characteristics shift markups and competitive pressure, hence prices
- Limitation: product characteristics may be endogenous

GIV / covariance restrictions (e.g., Gabaix and Koijen, 2019; MacKay and Miller, 2017)

- Use restrictions on the covariance of demand and supply shocks
- Useful when conventional cost shifters are unavailable
- Limitation: credibility depends on the economic restrictions

Elasticity of Demand

The elasticity of demand is given by

$$\frac{\frac{\partial s_{jt}}{\partial p_{jt}}}{\frac{s_{jt}}{p_{jt}}} = \alpha(1 - s_{jt})p_{jt}$$

And the cross-price elasticity of demand is

$$\frac{\frac{\partial s_{jt}}{\partial p_{kt}}}{\frac{s_{jt}}{p_{kt}}} = -\alpha s_{kt} p_{kt}$$

Note that these patterns could be restrictive and unrealistic in many settings.

Limitations

The model imposes some restrictive substitution patterns

Example: Red Bus/Blue Bus Problem

- Consumers commute by car or red bus.
- Suppose the car and red bus have equal utility, $u_C = u_R$.

$$s_C = \frac{\exp(u_C)}{\exp(u_R) + \exp(u_C)} = \frac{1}{2}.$$

- Now introduce an identical blue bus, $u_B = u_R$.
- Intuition: car share should remain 50 percent; bus riders split between red and blue buses.

$$s_C = \frac{\exp(u_C)}{\exp(u_R) + \exp(u_C) + \exp(u_B)} = \frac{1}{3}.$$

- Logit treats the blue bus as a genuinely new source of idiosyncratic utility, not as a duplicate of the red bus.

Limitations/Other Considerations

- Only source of consumer heterogeneity is ε_{ijt}
- The model implies consumers have a strong preference for variety
  - Consumers always benefit from new products because of the utility shock
  - i.e. the model implies consumers benefit from having the option to ride a red and a blue bus, even though consumers are indifferent between the two.

Random Coefficients Logit (BLP)

BLP (1995) allows preferences to vary across consumers.

$$u_{ij} = x_j' \beta_i + \xi_j + \varepsilon_{ij}.$$

- Consumers differ in their tastes for product characteristics.
- Similar products become closer substitutes.
- Price elasticities vary across consumers and products.
- The simple logit model is a special case.

Goal: Relax the unrealistic substitution patterns implied by the simple logit model while remaining computationally tractable.

Modeling Preference Heterogeneity

One way to parameterize consumer-specific preferences is

$$\beta_{ik} = \beta_k + \beta_k^d d_i + \beta_k^u v_{ik},$$

where

- d_i is an observed demographic characteristic (inc., wealth, age, etc.).
- v_{ik} is an unobserved taste shock.
- Taste shocks are typically assumed to satisfy

$$v_i \sim N(0, \Omega).$$

Observed demographics and unobserved tastes both generate heterogeneity in willingness to pay.

Mean Utility and Individual Deviations

Rewrite utility as

$$u_{ij} = \delta_j + \mu_{ij} + \varepsilon_{ij},$$

where

$$\delta_j = x_j'\beta + \xi_j,$$

and

$$\mu_{ij} = x_j'\beta^d d_i + x_j'\beta^u v_i.$$

This decomposition separates average demand from consumer-specific deviations.

- a common component (δ_j),
- consumer-specific tastes (μ_{ij}), and
- the familiar logit shock (ε_{ij}).

Mean Utility and Individual Deviations

Mean utility (δ_j)

- Common to all consumers
- Captures average demand
- Includes unobserved quality ξ_j
- Recovered from observed market shares

Individual deviation (μ_{ij})

- Varies across consumers
- Depends on demographics and unobserved tastes
- Generates richer substitution patterns
- Determines heterogeneous elasticities

Market Shares in BLP

Given δ and the heterogeneity parameters

$$\theta_2 = (\beta^d, \beta^u, \Omega),$$

market shares are obtained by averaging individual choice probabilities across consumers:

$$s_j(\delta, \theta_2) = \int \frac{\exp(\delta_j + \mu_{ij})}{1 + \sum_{\ell=1}^{J} \exp(\delta_\ell + \mu_{i\ell})} \, dF(d_i, v_i).$$

- The integral averages over the population distribution of tastes.
- We no longer have a closed-form expression because we must integrate over heterogeneous consumers.
- There is generally no closed-form solution.
- We approximate the integral using simulation.

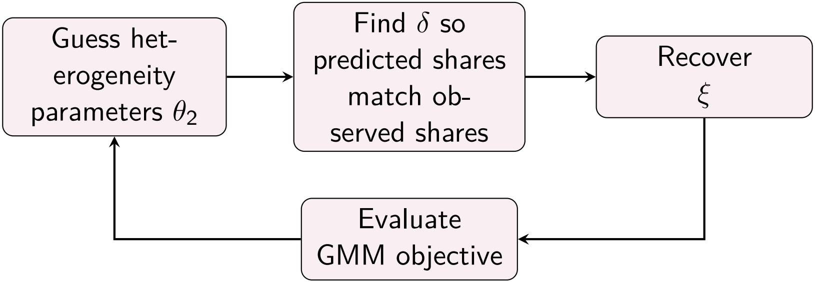
BLP Estimation: Overview

1. Draw a simulated population of consumers.
   - demographics d_i
   - taste shocks v_i
2. Guess the heterogeneity parameters θ_2.
3. Solve for the mean utilities $\delta(\theta_2)$ that match observed market shares.
4. Recover the implied unobserved demand shocks

$$\xi = \delta - X\beta.$$

5. Evaluate the GMM objective.
6. Update θ_2 and repeat.

BLP Estimation: Big Picture



The inner loop finds the mean utilities that rationalize observed market shares. The outer loop searches for the heterogeneity parameters that satisfy the moment conditions.

Inner Loop: Contraction Mapping

Holding θ_2 fixed, solve for δ such that

$$s^{obs} = s^{model}(\delta, \theta_2).$$

The standard update is

$$\delta_j^{(r+1)} = \delta_j^{(r)} + \ln s_j^{obs} - \ln s_j^{model}(\delta^{(r)}, \theta_2).$$

- If predicted demand is too low, increase δ_j.
- If predicted demand is too high, decrease δ_j.
- Iterate until predicted and observed shares match.

The contraction mapping simply recovers the mean utilities that rationalize the observed market shares.

Outer Loop: GMM

Given $\delta(\theta_2)$, recover

$$\xi(\theta) = \delta(\theta_2) - X\beta.$$

Choose the heterogeneity parameters so that

$$E[Z'\xi(\theta)] = 0.$$

A typical GMM objective is

$$\min_{\theta} g(\theta)' W g(\theta), \qquad g(\theta) = Z'\xi(\theta).$$

The estimator searches for the preference parameters that make the recovered demand shocks orthogonal to the instruments.

BLP Estimation: Putting It All Together

Inner loop

1. Guess the heterogeneity parameters θ_2.
2. Simulate market shares.
3. Solve for δ using the contraction mapping.

Outer loop

4. Recover $\xi = \delta - X\beta$.
5. Evaluate the GMM objective.
6. Update θ_2 and repeat.

Key idea:
The inner loop matches observed market shares. The outer loop chooses the preference parameters that satisfy the moment conditions.

Why use BLP?

What Random Coefficients Buy You

- Price elasticities vary across consumers and products.
- Substitution depends on product characteristics rather than only market shares.
- Similar products become close substitutes.
- Rich counterfactuals: mergers, entry, product design, and welfare analysis.

Example: If two ETFs track the same index, investors with a strong preference for that index will view them as close substitutes. Random coefficients capture this; the simple logit model cannot.

BLP is most valuable when realistic substitution patterns are central to the economic question.

Useful Resources

- Nevo (2000): "A Practitioner's Guide To Estimation of Random-Coefficients Logit Models of Demand"
    - Code available online
- Petrin (2002)
    - Quantifying the benefits of new products
- Dube, Fox, and Su (2012)
    - Use MPEC to estimate BLP
    - More coding work upfront, but can be faster
    - Code available online
- Goldszmidt et al. (2026)
    - Price elasticities in mixed logit models can be highly stylized
    - Particularly true when market shares are small
- Conlon and Gortmaker (2020, 2025)
    - PyBLP package
    - Best practices for BLP demand estimation

Examples of Demand Estimation in Finance

- Bank Deposits
  - Dick (2007)
  - Egan, Hortacsu and Matvos (2017)
  - Xiao (2020)
  - Xiao, Wang, Whited and Wu (2020)
  - Xiao, Whited, and Wu (2023)
- Brokers
  - Di Maggio, Egan, and Franzoni (2019)
- Mortgages
  - Benetton (2018)
  - Garcia (2019)
- Insurance
  - Koijen and Yogo (2020)
- Asset Pricing
  - Koijen and Yogo (2019)
  - Egan, MacKay, and Yang (2023; 2025)